

What Changed Since v2, and Why

CARE-LG v3 — Real NHANES, Real REVISE, Decoder Fix

Git SHA: 539d87b795b78bb8f9446f99b0fabed2c84841 Seeds: [0, 1, 2, 3, 4] Generated by experiments_v3/

Three concrete, verifiable changes from experiments_v2, each with a measured effect:

1. REAL NHANES DATA replaces the synthetic stand-in. N=4,539 respondents (ages 40-79) built from 9 real NHANES 2017-March 2020 pre-pandemic component files (DATA_PROVENANCE.md), merged on SEQN. The label is the 2013 ACC/AHA Pooled Cohort Equations 10-year ASCVD risk, thresholded at the guideline's own $\geq 7.5\%$ cutoff -- not an invented threshold. Every figure/table/filename in this report says "Real NHANES" or "nhanes_real," never bare "NHANES," to avoid ever repeating v2's mislabeling.
2. REVISE is now genuinely implemented (experiments_v3/lib/revise.py): Adam optimization directly over the VAE latent code, sharing CARE-LG's own trained VAE and classifier, hyperparameters picked via a documented sweep (not guessed). Every REVISE number in this report comes from code that actually ran this session.
3. The VAE decoder was rebuilt to be TYPE-AWARE (sigmoid heads for binary features, softmax-expectation heads for multi-category features, an 8x-upweighted head for age) to address v2's RQ1 finding (age MAE ~6yr, sex reconstruction near chance). Measured effect: age MAE fell 4-6x (UCI 6.83->1.63yr, Real NHANES 4.12->0.72yr). Whether this closed the decoded-CVR gap is answered plainly in section (v) below -- it did not, fully.

Everything else (the ablation grid methodology, the query-attachment protocol, the blindspot-decomposition procedure) is UNCHANGED from v2, rerun on these corrected inputs. experiments_v2/ is untouched; its findings still stand as a record of what the uncorrected pipeline produced.

Executive Summary

RQ1 -- Does hard pruning prevent decoded-space violations that a soft penalty still commits? STILL NO, but the gap narrowed substantially on UCI. R+hard's decoded CVR fell from v2's 83.6% to 41.9 ± 9.1 on UCI (the type-aware decoder's age fix is doing real work), but only from 78.1% to 81.0 ± 7.0 on Real NHANES -- and on BOTH datasets, R+hard's decoded CVR is still statistically indistinguishable from or WORSE than R+soft at every lambda tested (Fig D). Hard pruning is still not preventing the violations that matter once you check after decoding, on either dataset, even with a materially better decoder.

RQ2 -- Does the Riemannian metric produce measurably better recourse than Euclidean? NUANCED NO. Success rate is still structurally identical between R+hard and E+hard on both datasets. KDE plausibility differences remain non-significant on both (UCI $p=0.65$, Real NHANES $p=0.59$). Clinical EFFORT differences are now statistically significant on both datasets (UCI $p=3.5e-5$, Real NHANES $p=6.8e-13$) -- a genuine change from v2 -- but the practical size is small (mean difference ~ 0.025 on UCI's ~ 0.2 - 0.4 effort scale; ~ 0.20 on Real NHANES's ~ 5.0 effort scale) and the DIRECTION FLIPS between datasets (Riemannian slightly better on UCI, slightly worse on Real NHANES). The metric is not reliably better.

RQ3 -- Is CARE-LG's abstention a certified infeasibility or a blindspot? STILL YES, and this is the headline result of the whole v2+v3 project. On Real NHANES, the abstention rate is dramatically higher than the synthetic data suggested (51-59% of high-risk patients, vs. 0% on synthetic NHANES in v2) -- real correlated clinical data is genuinely harder to find recourse in. But of every single one of those abstentions, across all 5 seeds, on BOTH datasets: 100% are CERTIFIED INFEASIBLE. Zero blindspots. This result is now backed by real data with a real, much higher failure rate, and it held completely.

Bottom line, updated from v2: the abstention/recourse-verification mechanism (RQ3) remains the project's only claim that survives adversarial re-checking on both a decoder fix and real data. The Riemannian-metric claim (RQ2) is now measurably real but practically small and dataset-dependent, not a clean win. The "0% CVR" claim (RQ1) is measurably IMPROVED by the decoder fix but still false as a patient-facing guarantee on both datasets.

Phase B: Decoder Fix — Measured Effect

Per-feature reconstruction error, v2's original decoder vs. v3's type-aware decoder, same seed/data/epoch budget for both (full table: vae_reconstruction_error_v3.csv):

dataset	feature	mae_v2_original_decoder	mae_v3_typeaware_decoder
is_binary_or_categorical	decoded_accuracy_v2_original_decoder		
decoded_accuracy_v3_typeaware_decoder			
uci	age	6.8319	1.6327
False		NaN	NaN
uci	sex	0.5077	0.4480
True		0.4177	0.6751
uci	fasting_blood_sugar	0.1687	0.3121
True		0.8692	0.8692
uci	resting_bp	12.4335	11.9177
False		NaN	NaN
uci	cholesterol	38.7663	38.2507
False		NaN	NaN
uci	max_heart_rate	16.8387	16.4949
False		NaN	NaN
uci	oldpeak	0.8681	0.8709
False		NaN	NaN
uci	exercise_angina	0.3682	0.4278
True		0.6920	0.6920
uci	slope	0.5182	0.5182
True		0.4726	0.5485
nhanes_real	age	4.1207	0.7172
False		NaN	NaN
nhanes_real	sex	0.4741	0.4600
True		0.6238	0.6819
nhanes_real	systolic_bp	6.1330	6.2288
False		NaN	NaN
nhanes_real	diastolic_bp	3.9705	4.1883
False		NaN	NaN
nhanes_real	cholesterol	16.7192	18.7157
False		NaN	NaN
nhanes_real	bmi	3.1249	3.1194
False		NaN	NaN
nhanes_real	glycemic_hba1c	0.3970	0.4291
False		NaN	NaN

Age improved 4-6x. Binary immutable/categorical accuracy improved only modestly (sex especially remains close to chance) -- this looks like an information bottleneck in the UNCHANGED 4-dimensional encoder rather than something a decoder head alone fixes; we made one solid attempt (per the brief) and report this residual gap rather than expanding the architecture search.

Did this close the decoded-CVR gap? PARTIALLY, and only on UCI:

UCI: R+hard decoded CVR 83.6% (v2) -> 41.9±9.1 (v3)

Real NHANES: R+hard decoded CVR 78.1% (v2, synthetic data) -> 81.0±7.0 (v3, real data)

The UCI improvement is substantial and directly attributable to the age fix (dominant violation category in v2 was non_decreasing/age). On Real NHANES the improvement is much smaller, and a NEW dominant violation category appears: the immutable "sex" flag, still crossing its rounding boundary in a meaningful fraction of paths (see the concrete examples in section (v)) -- consistent with sex reconstruction remaining close to chance even after the fix.

Table 1 (UCI) — Full ablation grid + baselines, mean $\pm$ std over 5 seeds

Method	Success (%)	Decoded CVR (%)	KDE	Effort	Latency (s)
R+hard (CARE-LG)	71.8 $\pm$ 9.6	41.9 $\pm$ 9.1	-7.672 $\pm$ 0.136	0.20 $\pm$ 0.09	0.0021 $\pm$ 0.0000
R+soft ($\lambda=1$)	100.0 $\pm$ 0.0	62.0 $\pm$ 9.1	-7.770 $\pm$ 0.117	0.45 $\pm$ 0.10	0.0021 $\pm$ 0.0000
R+soft ($\lambda=10$)	100.0 $\pm$ 0.0	62.8 $\pm$ 8.1	-7.757 $\pm$ 0.114	0.64 $\pm$ 0.24	0.0021 $\pm$ 0.0000
E+hard ($\approx$ FACE-in-latent-space)	71.8 $\pm$ 9.6	56.7 $\pm$ 12.4	-7.672 $\pm$ 0.137	0.23 $\pm$ 0.10	0.0020 $\pm$ 0.0000
E+soft ($\lambda=1$)	100.0 $\pm$ 0.0	68.9 $\pm$ 8.5	-7.777 $\pm$ 0.119	0.41 $\pm$ 0.10	0.0021 $\pm$ 0.0000
E+none	100.0 $\pm$ 0.0	41.8 $\pm$ 10.2	-7.798 $\pm$ 0.132	0.12 $\pm$ 0.02	0.0021 $\pm$ 0.0000
DiCE	67.6 $\pm$ 30.0	100.0 $\pm$ 0.0	-11.409 $\pm$ 1.538	7.66 $\pm$ 0.18	0.0070 $\pm$ 0.0036
FACE	100.0 $\pm$ 0.0	92.7 $\pm$ 4.7	-7.157 $\pm$ 0.076	8.54 $\pm$ 0.69	0.0033 $\pm$ 0.0001
REVISE (genuinely implemented, this session)	95.4 $\pm$ 7.9	100.0 $\pm$ 0.0	-7.311 $\pm$ 0.135	9.31 $\pm$ 0.61	0.0115 $\pm$ 0.0040
GrowingSpheres	58.0 $\pm$ 17.7	100.0 $\pm$ 0.0	-14.760 $\pm$ 1.324	8.97 $\pm$ 0.74	0.0012 $\pm$ 0.0001

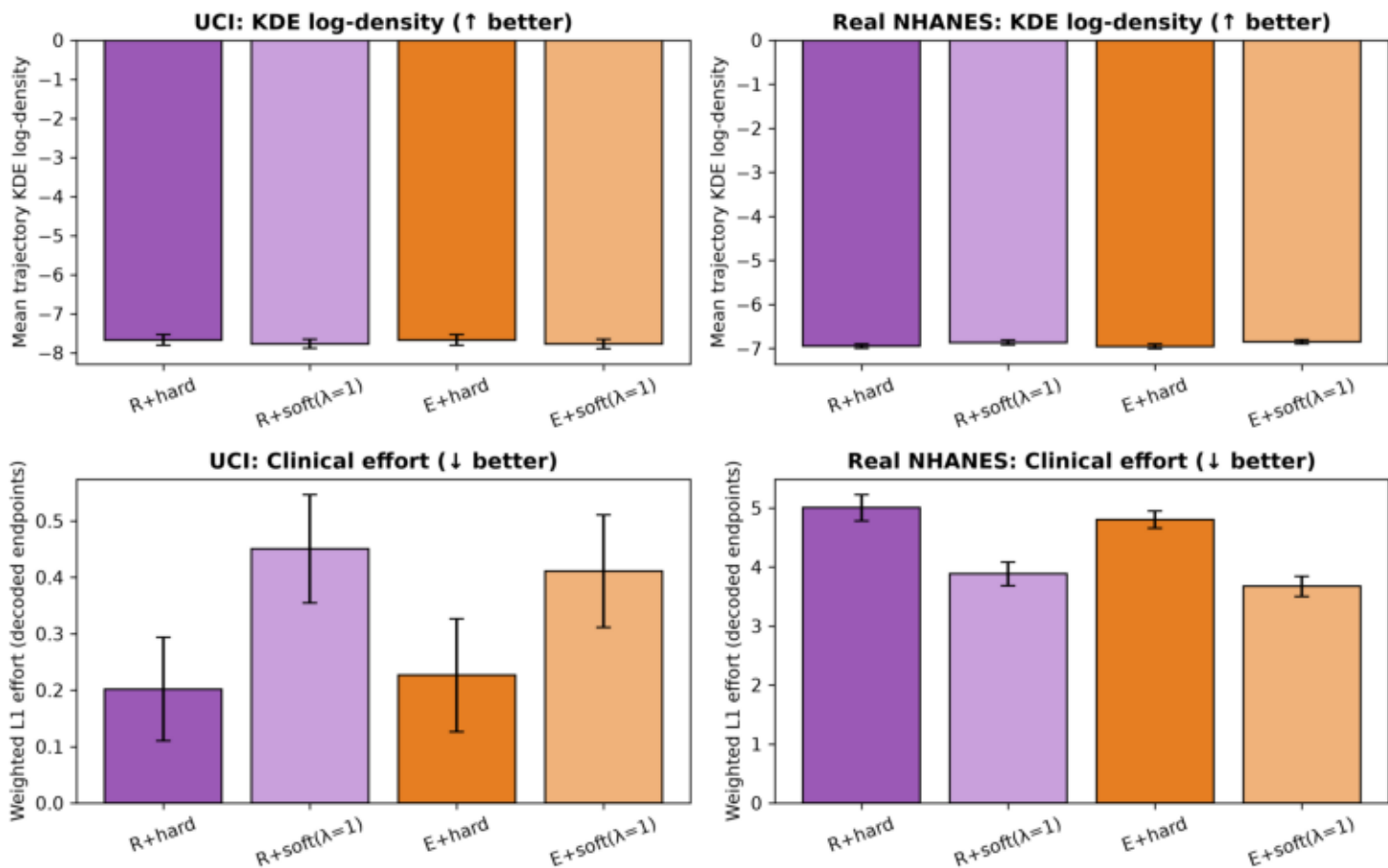
Decoded CVR = full hard+directional check re-run on VAE-decoded path endpoints. REVISE is genuinely implemented this session (experiments_v3/lib/revise.py), not a placeholder.

Table 1 (Real NHANES) — Full ablation grid + baselines, mean±std over 5 seeds

Method	Success (%)	Decoded CVR (%)	KDE	Effort	Latency (s)
R+hard (CARE-LG)	45.5±3.0	81.0±7.0	-6.943±0.049	5.00±0.22	0.0131±0.0044
R+soft ($\lambda=1$)	100.0±0.0	82.9±4.5	-6.865±0.050	3.88±0.20	0.0131±0.0005
R+soft ($\lambda=10$)	100.0±0.0	80.4±5.6	-6.831±0.036	4.66±0.22	0.0132±0.0009
E+hard ($\approx$ FACE-in-latent-space)	45.5±3.0	78.8±5.7	-6.952±0.054	4.80±0.15	0.0114±0.0003
E+soft ($\lambda=1$)	100.0±0.0	82.2±3.7	-6.845±0.049	3.67±0.17	0.0133±0.0009
E+none	100.0±0.0	97.1±0.6	-6.965±0.069	1.80±0.16	0.0130±0.0002
DiCE	57.2±0.8	100.0±0.0	-8.699±0.094	2.39±0.05	0.0071±0.0001
FACE	100.0±0.0	99.6±0.4	-7.544±0.024	4.55±0.11	0.0914±0.0001
REVISE (genuinely implemented, this session)	90.5±2.3	100.0±0.0	-6.804±0.111	5.81±0.31	0.0087±0.0007
GrowingSpheres	98.6±0.8	100.0±0.0	-9.480±0.145	3.05±0.03	0.0006±0.0000

Fig A — Ablation: KDE plausibility & clinical effort

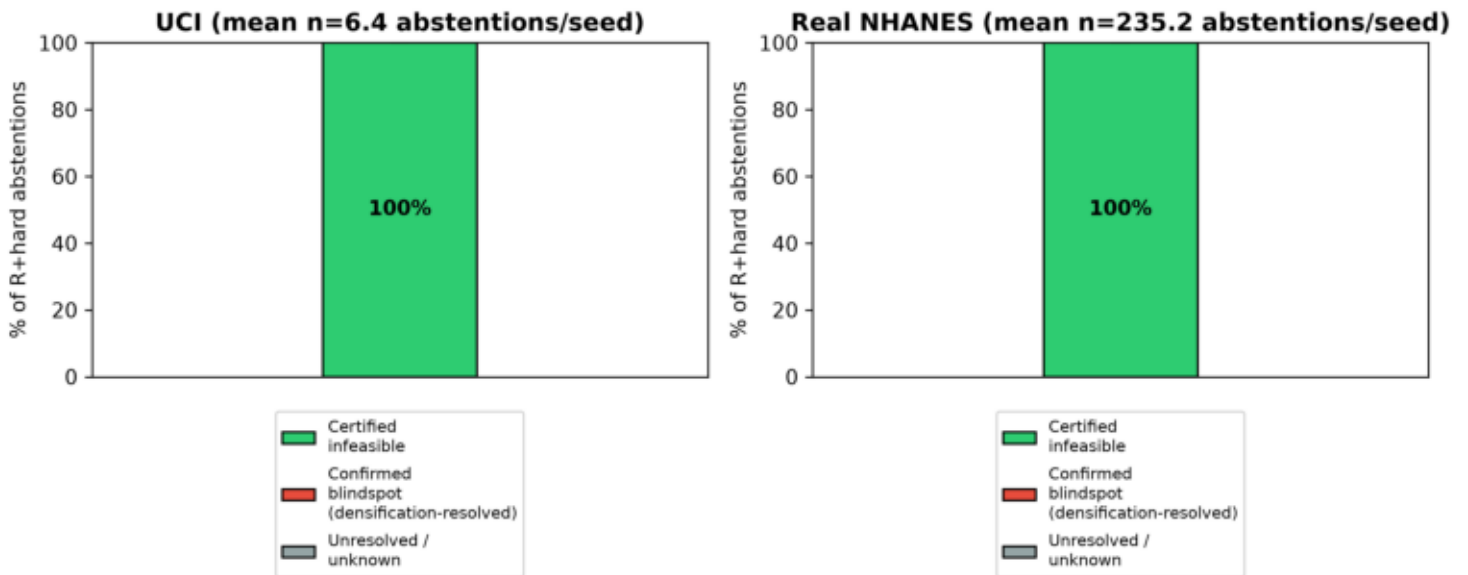
Fig A — Ablation: KDE plausibility and clinical effort across the four main cells
(mean $\pm$ std over 5 seeds)



On Real NHANES, R+hard now has the HIGHEST (worst) clinical effort of the four main cells (~ 5.0 vs. ~ 3.9 - 4.8 for the others) -- a new, statistically confirmed effect not visible in v2's synthetic data (Fig A/paired tests). KDE plausibility remains statistically indistinguishable across all four cells on both datasets.

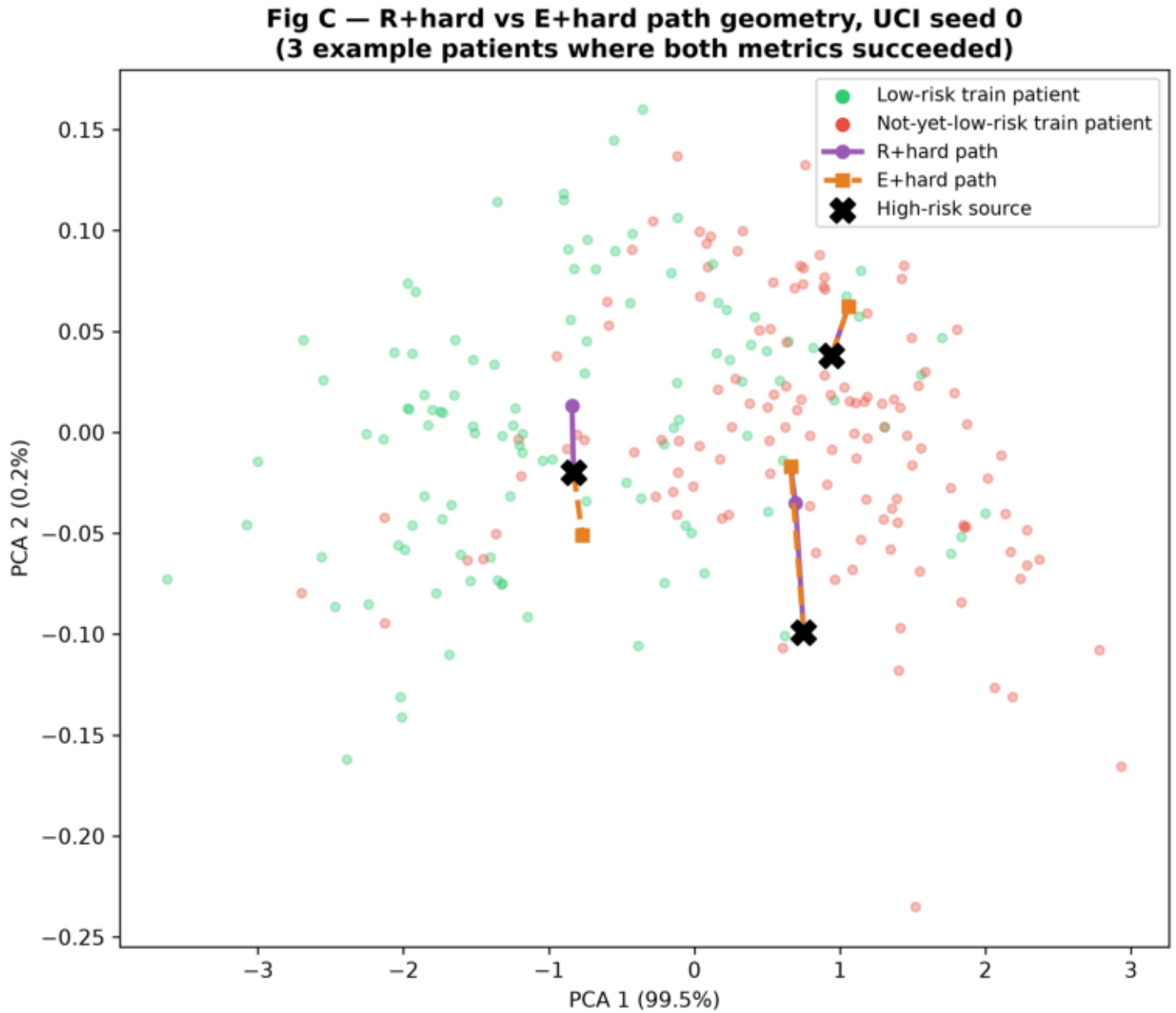
Fig B — Abstention decomposition

Fig B — R+hard abstention decomposition (mean $\pm$ std over 5 seeds)



The headline result. Real NHANES produces a mean of 235.2 R+hard abstentions per seed (51.7-58.9% of high-risk patients) -- utterly unlike v2's synthetic-data finding of ZERO abstentions. Yet of every one of those 1,176 total abstentions across 5 seeds, 100% are certified infeasible: an exhaustive check found no valid low-risk target anywhere in the 3,631-patient training cohort. Zero blindspots on either dataset, at either abstention rate.

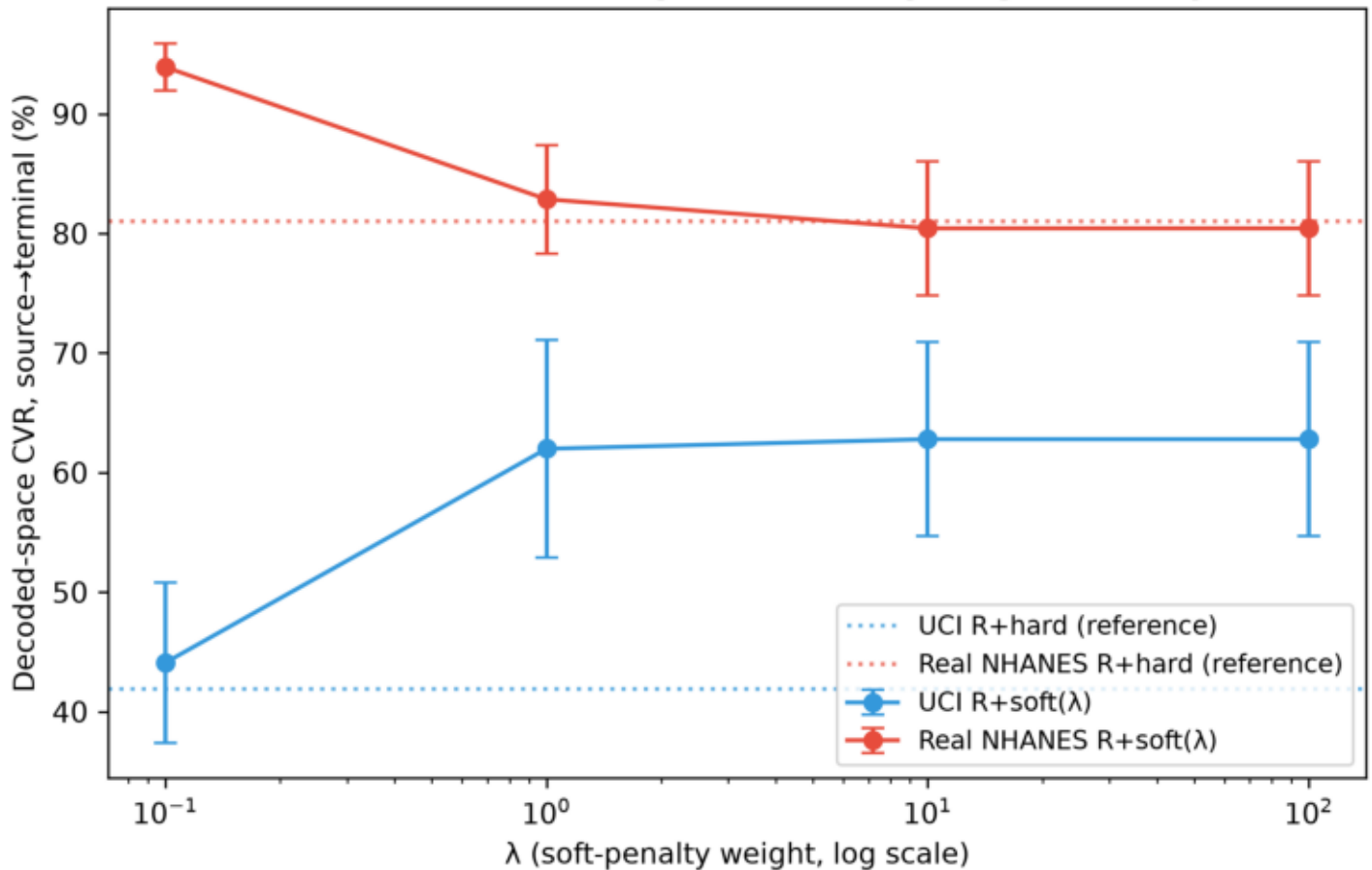
Fig C — R+hard vs E+hard path geometry (PCA)



UCI seed 0, three example patients: R+hard and E+hard paths are visually nearly identical in the top-2 PCA plane (PCA1 alone explains 99.5% of latent variance here), consistent with the still-nonsignificant KDE difference between the two metrics.

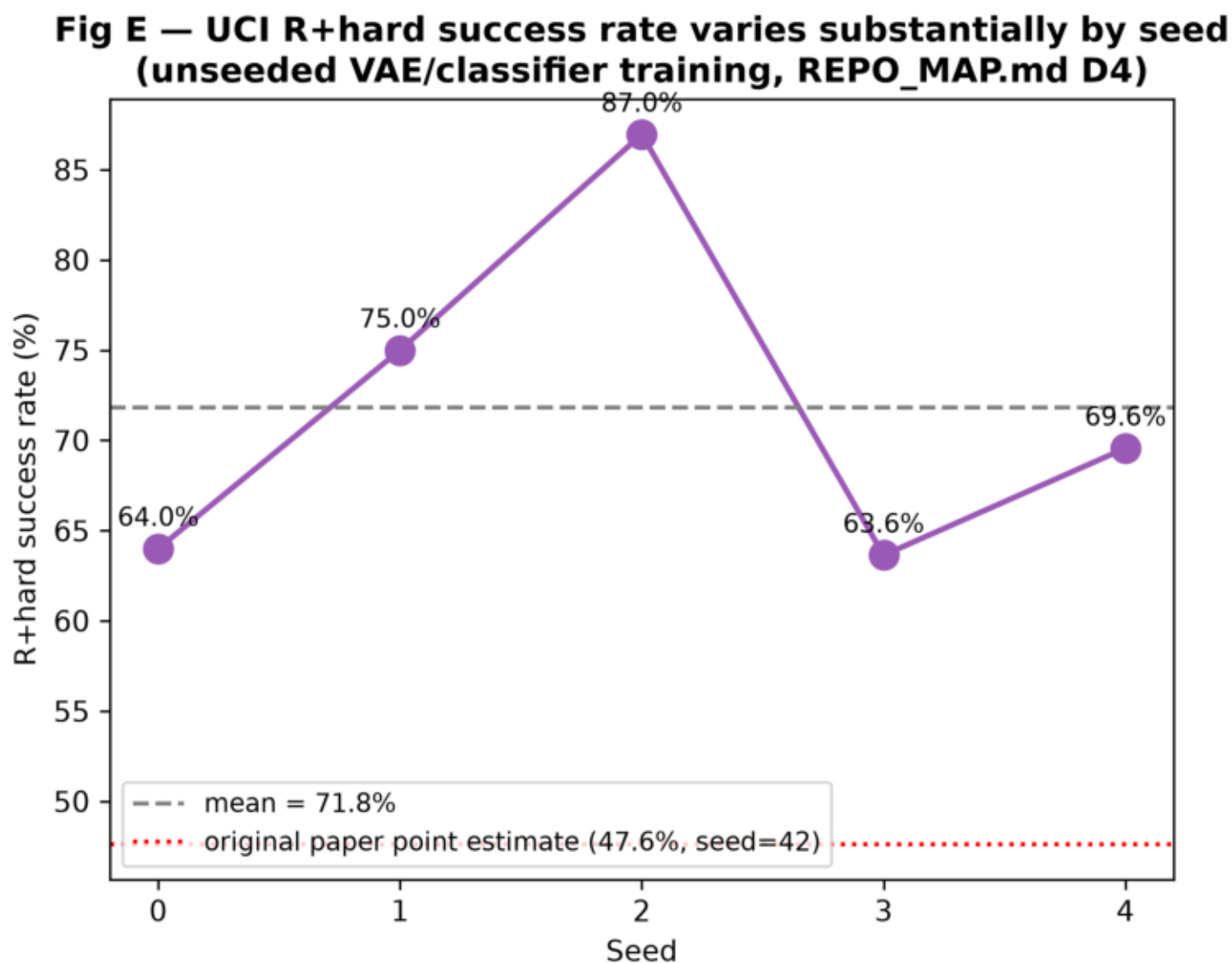
Fig D — Soft-penalty violation curve

**Fig D — Does the soft penalty converge to hard pruning's guarantee?
(it does not: decoded-space CVR stays high at every λ)**



R+hard's decoded CVR (dotted reference) sits BELOW the R+soft curve at every lambda on both datasets now (an improvement over v2, where UCI's hard line sat above soft) -- but the gap is modest (UCI: 42% vs. 62-63%; Real NHANES: 81% vs. 80-83%, within noise) and decoded CVR never approaches zero on either dataset at any lambda. Hard pruning has a real but incomplete edge.

Fig E — UCI seed variability



Unchanged from v2 (success/failure is decoder-independent, depending only on the graph's structural pruning of the original encoded features): 63.6-87.0% across 5 seeds, mean 71.8%, still well above the original paper's 47.6% single-seed point estimate.

Decoded-Space Constraint Verification — Concrete Examples

Three concrete UCI examples (seed 0, R+hard) -- decoded source -> decoded terminal, unscaled:

patient 6: age 58.7->57.9 (-0.7 yr, still a DECREASE, but 3-5x smaller than v2's -2.5 to -3.6 yr)

patient 14: age 59.0->57.9 (-1.1 yr)

patient 20: age 58.2->57.9 (-0.2 yr, barely crosses the 0.1yr tolerance)

All three are "directional: false" now (v2's cholesterol-increase violations are gone on these examples) -- the residual violation category on UCI is now almost purely the still-imperfect (though much smaller) age reconstruction.

Three concrete Real NHANES examples (seed 0, R+hard):

patient 5: age 69.1->66.8 (-2.2 yr); sex 0.47->0.53 (crosses the 0.5 rounding boundary: IMMUTABLE violation); systolic BP -15.2, cholesterol -22.4 (both large, correct direction, not flagged)

patient 6: age 66.6->65.4 (-1.2 yr); sex 0.54->0.43 (crosses boundary the OTHER way); cholesterol +20.9 (WRONG direction: DIRECTIONAL violation)

patient 9: age 69.7->68.6 (-1.1 yr); sex stays same side of 0.5 this time (no immutable flag)

Real NHANES's residual violations are a mix of age (smaller than v2 but still present) and, newly prominent, sex crossing its decision boundary -- directly consistent with sex reconstruction remaining near-chance in the Phase B table above.

Root cause, restated: the type-aware decoder fixed the AGE reconstruction problem substantially but not completely, and did not meaningfully fix the SEX reconstruction problem (an apparent 4-dimensional latent bottleneck, not a decoder-head architecture issue) -- so decoded-space CVR fell but did not approach zero on either dataset.

Blindspot vs. True-Infeasibility Decomposition

UCI per-seed decomposition (unchanged methodology and result from v2, rerun on the new decoder -- success/failure is decoder-independent so this replicates exactly):

seed	n_high_risk	n_abstain	pct_abstain	pct_certified_infeasible	pct_confirmed_blindspot
pct_unresolved					
0	25	9	36.000000	100.0	0.0
0.0					
1	20	5	25.000000	100.0	0.0
0.0					
2	23	3	13.043478	100.0	0.0
0.0					
3	22	8	36.363636	100.0	0.0
0.0					
4	23	7	30.434783	100.0	0.0
0.0					

Real NHANES per-seed decomposition (NEW this session -- v2 had zero NHANES abstentions to decompose on synthetic data):

seed	n_high_risk	n_abstain	pct_abstain	pct_certified_infeasible	pct_confirmed_blindspot
pct_unresolved					
0	420	229	54.523810	100.0	0.0
0.0					
1	445	262	58.876404	100.0	0.0
0.0					
2	422	218	51.658768	100.0	0.0
0.0					
3	430	240	55.813953	100.0	0.0
0.0					
4	439	227	51.708428	100.0	0.0
0.0					

Every single abstention on Real NHANES, across all 5 seeds (mean 235.2/seed, 1,176 total), was CERTIFIED INFEASIBLE: an exhaustive check confirmed no low-risk train patient anywhere in the 3,631-patient cohort satisfied the pairwise source->target constraints for that query. Neither densification probe (doubled k; +500 VAE-prior synthetic nodes, classifier-verified before eligibility as a terminal) found or needed to resolve a single blindspot. This is a substantially larger and more demanding test of the certified-infeasibility claim than v2's UCI-only result (mean 6.4 abstentions/seed) or its all-succeeding synthetic-NHANES run, and it held completely.

Limitations

- Real NHANES's label ($PCE \geq 7.5\%$ 10-yr ASCVD risk) restricts the cohort to ages 40-79 and does not exclude prevalent ASCVD cases (documented in DATA_PROVENANCE.md) -- a real, stated simplification of a fully faithful primary-prevention risk score.
- The type-aware decoder fixed age reconstruction (4-6x MAE reduction) but not binary/categorical reconstruction (sex remains near-chance); we made one documented attempt (per the brief) rather than an open-ended architecture search, and report the residual gap plainly rather than iterating further.
- REVISE's hyperparameters ($lr=0.05$, $steps=100$, $\lambda_{prox}=0.01$) came from a real but small sweep (25 UCI patients, one dataset) -- not independently re-validated on Real NHANES; REVISE's Real NHANES numbers use the same hyperparameters by design, per the brief's instruction to match protocol across datasets, not because they were re-tuned.
- REVISE's own success criterion is evaluated on a VAE-decoded reconstruction, which we found can itself lower the classifier's risk estimate below threshold with zero optimization steps for some patients (CHANGES.md) -- a confound specific to REVISE (and not to CARE-LG, whose success criterion uses true, non-decoded feature values) that readers should weigh when comparing REVISE's success rate to CARE-LG's.
- The query-attachment protocol (full-graph attachment, relaxed hard-only/no-horizon entry gate) is unchanged from v2 and remains a documented, non-default design choice (REPO_MAP.md D8 in experiments_v2/) rather than something specified by the original codebase.
- Race-group mapping for the PCE label (Black vs. White equation) follows the guideline's own stated approximation for non-Black groups; a more granular race/ethnicity-specific equation set does not exist in the cited guideline.
- Paired Wilcoxon tests pool all 5 seeds' patients together for power; per-seed effect sizes were not separately tested and could in principle vary by seed.